5. **In this question you should show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

Using the substitution $u = 3 + \sqrt{2x - 1}$ find the exact value of

$$\int_{1}^{13} \frac{4}{3 + \sqrt{2x - 1}} \, dx$$

giving your answer in the form $p + q \ln 2$, where p and q are integers to be found.

(8)